FROM WEAK SIGNALS TO PROACTIVE DECISIONS

How can AI help ABB see market changes before they happen?

ABB Distribution Solutions | Hackathon Challenge

The Core Challenge

How could ABB Distribution Solutions detect changes, fluctuations, and turning points in electrification markets earlier — and turn those findings into useful information that supports decisions in manufacturing, procurement, capacity planning, and supply reliability?

Global trend → market data → weak signal → scenario → business impact → decision

1. Background and Problem

The world is electrifying. More electricity is being produced, transmitted, distributed and consumed than ever before. This increases the demand for the protection and control solutions that ABB Distribution Solutions makes. However, this growth is not smooth or predictable.

Many factors affect the market: economic cycles, data center investments, changes in energy production, grid investments, geopolitical events, regional regulations, and component availability. ABB does not just need to know that electrification is growing. The real value comes from knowing where and when demand is speeding up, slowing down, or shifting — and acting on that knowledge before it shows up in actual orders.

The goal is not to predict exact sales figures. The goal is to build a model that catches early signals, explores different scenarios, and helps people make better decisions — before market changes become obvious.

2. Your Task

Choose a market or phenomenon related to electrification. Use public data and AI tools to explore it. Find the key drivers and uncertainties, identify weak signals and possible turning points, and think about how these could affect ABB's demand, manufacturing, procurement, or supply chain.

Based on your market and data analysis, define a specific decision problem that is relevant for ABB. Then design an AI-based digital solution that helps detect changes earlier, explore alternative scenarios, and support human decision-making.

Your team needs to show:

Which public data sources you can use and how to access them
How reliable, up-to-date, and forward-looking different signals are
How to separate long-term structural growth from short-term fluctuations and temporary disruptions
How to handle both growth-supporting and reversal-warning evidence in a balanced way
What alternative market scenarios can be built from the signals
What manufacturing, procurement, capacity, or supply chain decisions the findings could support
Why the AI solution you propose fits the problem and how it would work in practice

3. What to Deliver

Your final deliverable should combine business understanding, data analysis, AI solution logic, and a modern digital presentation. It should cover these six areas:

Area	What to include
1. Business & Market Insight	Key market drivers, growth and slowdown signals, regional differences, uncertainties, and the connection to ABB.
2. Data & Signal Portfolio	Data sources, leading and lagging indicators, data quality, how current the data is, lead time, and any usage restrictions.
3. Decision Problem	Who is the user, what decision they need to make, what change they need to detect, how much lead time is needed, and what happens if the signal is misread.
4. AI Solution Concept	System architecture, agent roles, step-by-step process, how evidence is handled, scenarios, and the role of human judgment.
5. Demonstrable Prototype	One end-to-end example that shows how the solution works and what value it delivers to the user.

Minimum requirement for your prototype:

New data → signal detection → source and counter-evidence review → scenario → decision recommendation for ABB → next action.

Here are some ideas for your presentation — you don't need to do all of these:

Interactive signal radar or market map
Filters by region, segment, and time horizon
Views showing signal strength, direction, and uncertainty
Scenario comparison and signal-based change paths
A combination of dashboard and chat interface
Automatically generated executive briefing or situation overview
Clickable prototype, lightweight live demo, or recorded use case video

4. Support and Resources Available

ABB will support you throughout the hackathon. Support is structured so that you start with business, market, and data understanding.

- Introduction to electrification megatrends and markets
- Basic knowledge of electrical systems and the role of protection and control solutions
- Overview of ABB Distribution Solutions' customer segments and operating environment
- Examples of how demand, manufacturability, procurement, and supply reliability are connected
- Mentoring from ABB experts

5. Evaluation Criteria

Teams will be evaluated on the following criteria:

Criteria	Weight
Business value and connection to ABB decisions	20 %
Understanding of the market and the phenomenon	20 %
Quality of data sources and signal chain	15 %
Quality of problem definition	15 %
Logic and clarity of the AI solution	20 %
Prototype functionality and modern presentation	10 %

The Hackathon Outcome — in One Sentence

An AI solution concept and modern prototype — grounded in business understanding, market knowledge, and public data — that detects weak signals and turning points in electrification markets and turns them into scenarios and recommended actions that support proactive decision-making at ABB.

Business-led, data-grounded and AI-enabled foresight for resilient decisions.